Kaustubh Hemant Sonawane

Bristol, UK | ksstubh.sonawane@gmail.com | +44 7388229743 | +91 8591441019 | www.github.com/Arishtanemi4 | www.linkedin.com/in/kaustubhson4 | https://kaustubh-sonawane-portfolio.vercel.app

UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Engineer with 3 years of industry experience at HDFC Asset Management taking LLM and NLP systems from prototype to production for financial-markets users: a daily pipeline that turns 300+ rating-agency emails into structured, entity-matched data, Azure OpenAI tool-calling agents (a WhatsApp bot used daily by roughly 50 to 100 Sales team members) and RAG applications on pgvector, delivered as Python FastAPI services in Docker on AWS with CI/CD. Fixed what broke in real use and supported each release after launch. MSc in Data Science at the University of Bristol (awaiting result) with an AstraZeneca dissertation. Seeking an AI Engineer role shipping reliable, inspectable LLM pipelines for market intelligence.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

LLM & NLP Pipelines in Production

- Built a **Python** pipeline that turns **300+ daily emails** from credit rating agencies into structured data: it parses each email, uses **fuzzy matching** to keep only the companies analysts track and sends a daily summary of previous and changed ratings, **saving 60 analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when an agency began sending emails as images.
- Built an **agentic WhatsApp fund-comparison bot** (via the Netcore API) used daily by the Sales team (roughly 50 to 100 users, 400 to 1,000 requests in working hours): **Azure OpenAI tool calling** picks the comparison type and extracts the fund names, then calls **Python functions** over SQL data and returns fund indicators and a chart PDF. Parameter extraction broke first; fixed with example inputs in the JSON tool definitions.
- Designed and built **RAG** applications end to end (**ReactJS**, **Python FastAPI** microservices, **LlamaIndex**, **Azure OpenAI GPT-3.5 & Llama-3**) for Compliance, Client Service and Investment, initially as the first engineer on the team alongside my manager; a separate **pgvector** database with **Azure OpenAI** embeddings per department kept access siloed.
- Took applications from prototype to real users, iterating on what broke; guardrailed behaviour through **system prompts**, versioned every prompt change in **Git** and had Compliance review each release; allocated models by cost, self-hosting **Llama-3** on a **GPU EC2** instance for the largest team.

Data Pipelines & Financial Data

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with legacy analyst data; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.
- Developed **Power BI** dashboards for the investment team's credit report analysis, improving efficiency by **10 hours a week**; recreated them in the ReactJS frontend for wider use without Power BI licensing costs.
- Worked directly with Investment, Compliance and Client Service stakeholders; **de facto team lead for up to 4 interns**; **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

MLOps, Deployment & Reliability

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS** (**EC2 including a GPU instance, S3, ECR and Lambda**), packaged the **FastAPI** microservices as **Docker** containers and automated deployment of the frontend and backend (dev and prod) through a one-click **Jenkins and GitLab CI/CD** pipeline.
- Ran both **real-time and batch workloads**: user-facing services on FastAPI microservices, and a daily rating-email pipeline run on **AWS SageMaker** and scheduled by an **EventBridge** rule that triggers a **Lambda** function at 9 AM.
- Implemented **RBAC** with **LDAP** authentication, restricted access to company private IPs with **AWS Security Groups** and deployed an **Nginx** reverse proxy, one of the fixes that took the applications from failing to passing the security team's **VAPT**.

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. My focus: the **scoring and ranking algorithm** across six evidence layers; the **data preprocessing pipeline** (14 clean, joined tables from 17 raw inputs, with integrity checks); and the **RAG-based AI assistant** that narrates ranked results and answers methodology questions. Returns a ranked shortlist of cancer cell lines with the evidence behind each pick, refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with **108 LLM personas** across 12 archetypes and a vision model that scores uploaded photos (ReactJS, Python FastAPI, NVIDIA NIM with the model selectable per request). LLM output treated as untrusted: **validated with Pydantic, repaired and cached by input hash**; rate limits shaped the design (12 calls instead of 108, results streamed). GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. **Multi-architecture deep learning study** comparing CNN, ResNet and ViT encoders trained from scratch with LSTM/attention decoders on three synthetic datasets across five seeds, evaluated with SBERT cosine similarity, BLEU and ROUGE-L and benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git

ACHIEVEMENTS

- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle credit card fraud detection problem.)
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework for AI-assisted coding) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2 incl. GPU, S3, ECR, Lambda, SageMaker, EventBridge), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, LDAP, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite